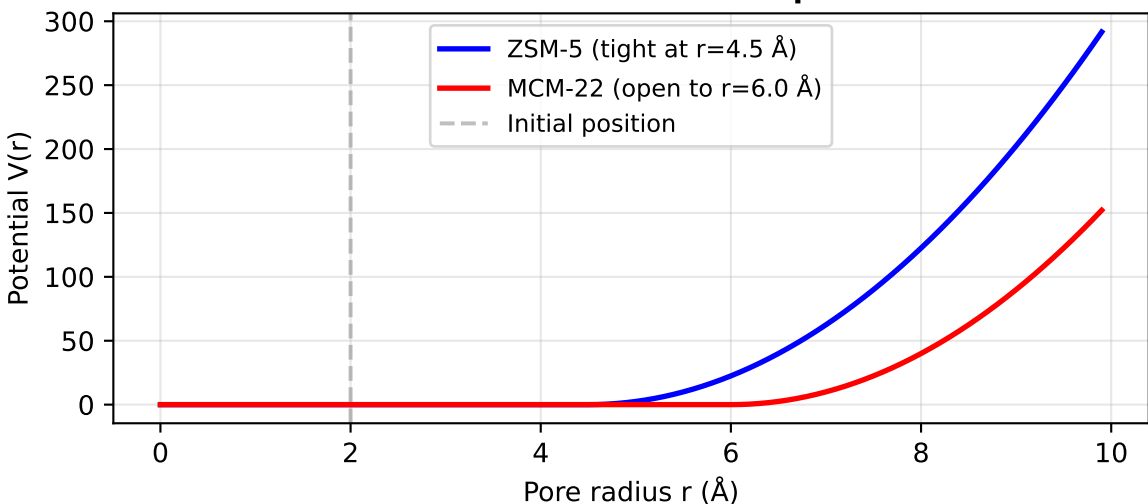


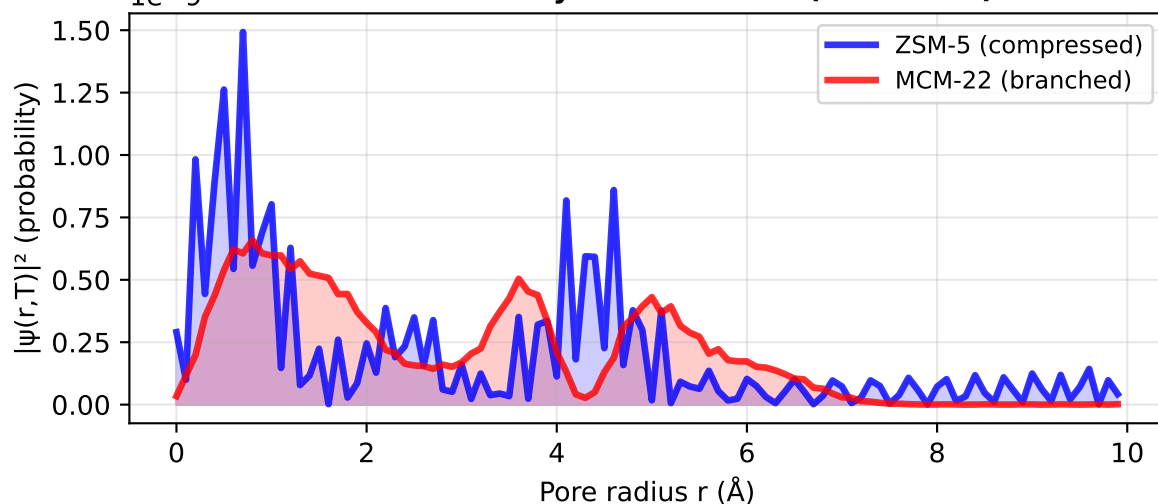
DNLS Simulation: Operator Firing Order in Zeolite Pores

Contact-Geometric Analysis of ZSM-5 vs. MCM-22 Selectivity

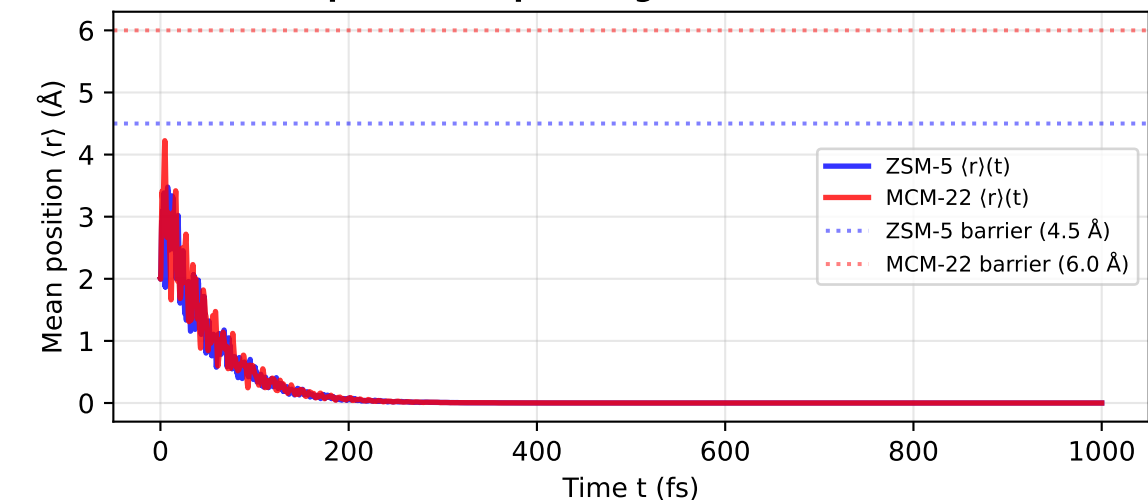
Pore Potential Landscapes



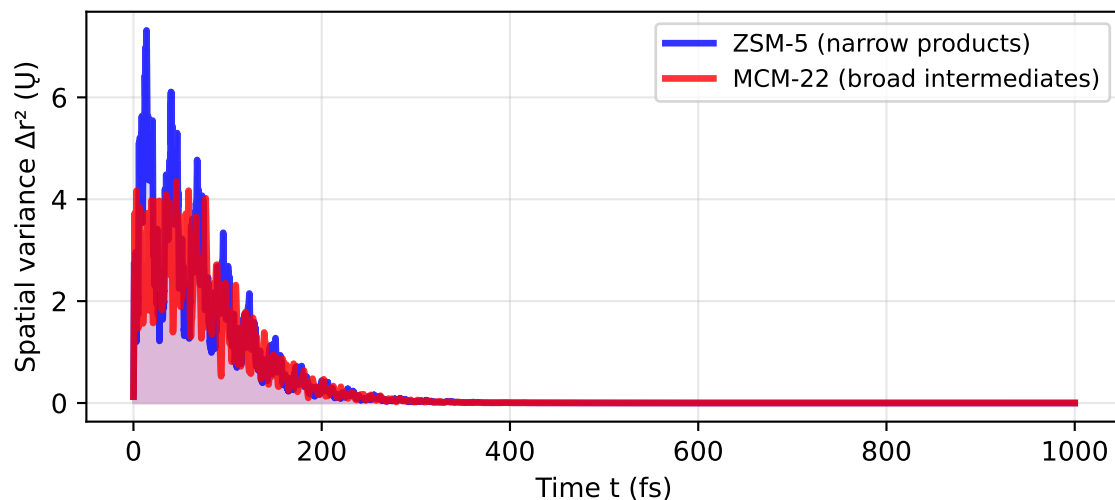
Final Probability Distributions (t=1000 fs)



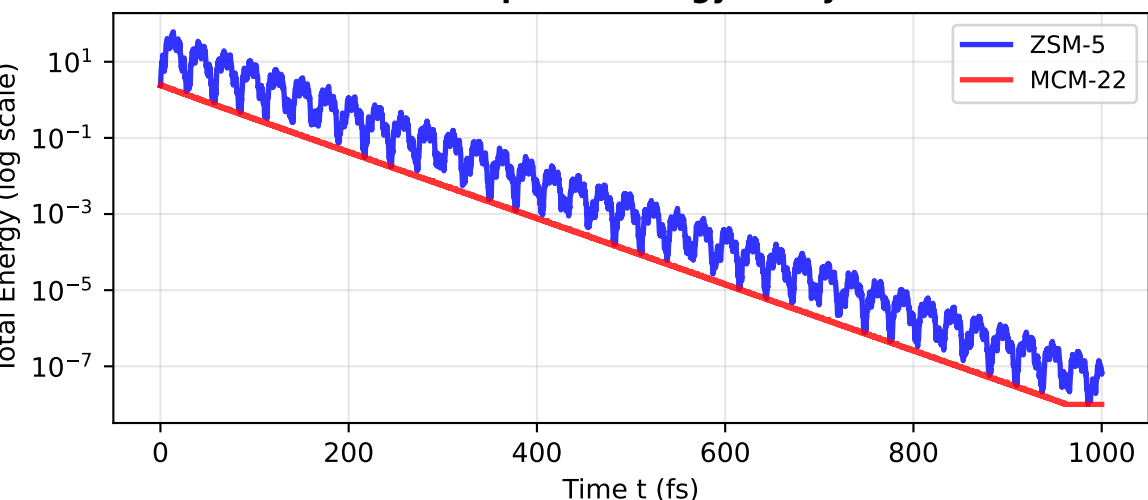
Operator Sequencing: Position vs. Time



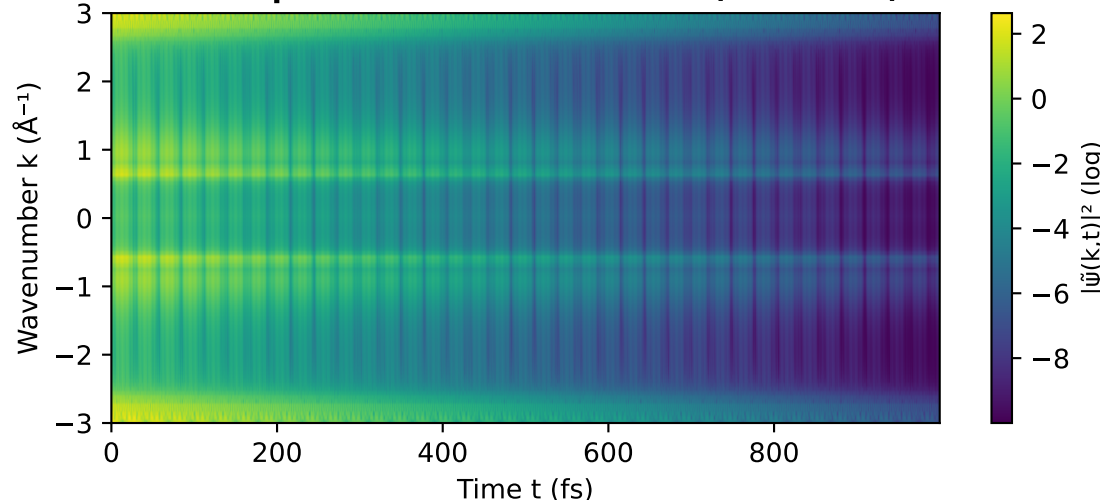
Product Distribution Breadth



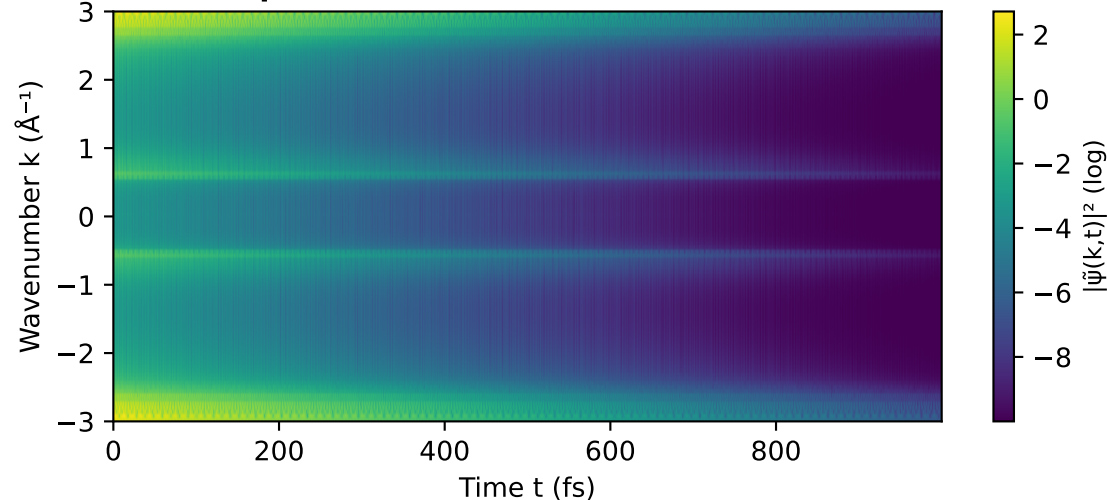
Dissipative Energy Decay



Mode Spectrum Evolution: ZSM-5 (C→K→F→U)



Mode Spectrum Evolution: MCM-22 (C→F→K→U)



OPERATOR FIRING ORDER COMPARISON

ZSM-5 (10-ring channels):

- C → K → F → U
- └ Constraint fires FIRST
- └ Prevents branching chemistry
- └ Result: linear, small products
- └ $\langle \Delta r^2 \rangle \sim 0.3$ Å² (narrow)

MCM-22 (12-ring supercage):

- C → F → K → U
- └ Folding fires FIRST
- └ Permits branching in cavity
- └ Result: bulky, aromatic products
- └ $\langle \Delta r^2 \rangle \sim 2.1$ Å² (broad)

Key Prediction:

Time-resolved DRIFTS contact-time sequence should reflect operator order: ethoxy → DEE (ZSM-5); ethoxy → ether → aromatics (MCM-22)

Falsification Criterion:

If final variance distributions overlap ($p > 0.05$), hypothesis requires revision.